44.2: An animal's nitrogenous wastes reflect its phylogeny and habitat

In regulating and safeguarding their internal fluids, animals must deal with **ammonia**, a toxic metabolite produced by the dismantling of *nitrogenous* (nitrogen-containing) molecules, chiefly proteins and nucleic acids. Several mechanisms have evolved for ridding the body of ammonia and other metabolic waste products, a process called **excretion**. Because most metabolic wastes must be dissolved in water to be excreted from the body, the type and quantity of an animal's waste products may have a large impact on its water balance.

Forms of Nitrogenous Waste

Although some animal species excrete ammonia directly, others excrete alternative forms of nitrogenous waste, either urea or uric acid **(Figure 44.6)**. These different forms vary significantly in their toxicity, their solubility, and the energy costs of producing them.

Figure 44.6

Variations in forms of nitrogenous waste among animal species.

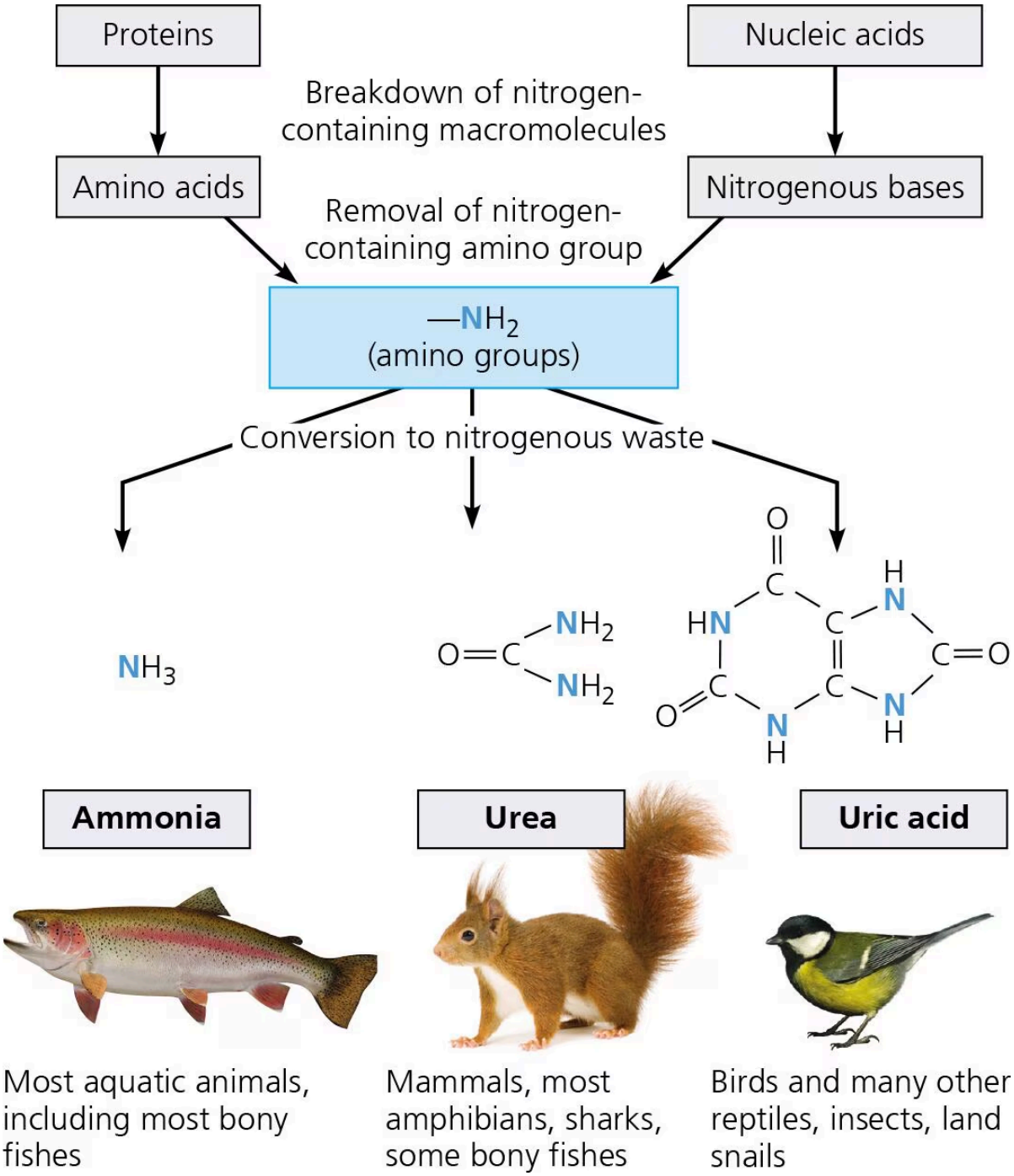

Chart depicting forms of nitrogenous waste including sources, animals that produce the waste, and chemical structures for the waste. All nitrogen atoms (N) are highlighted for emphasis. Proteins are broken down into amino acids; Nucleic acids are broken down into nitrogenous bases. Both contain amino groups (-N H 2). Forms of nitrogenous waste and animals that produce the include:

Ammonia: Most aquatic animals, including most bony fishes. Chemical structure is NH3.

Urea: Mammals, most amphibians, sharks and some bony fishes. Chemical structure is a carbonyl group (C double bonded to O) with the carbon attached to two amino groups (N H 2).

Uric Acid: Birds and many other reptiles, insects, land snails. Chemical structure consists of two rings (six sided and five sided), each containing two nitrogen atoms. There are two carbonyl groups on the six-sided ring and one on the five-sided ring.

Ammonia

Ammonia is very toxic, in part because its ion, ammonium NH_4^+, can interfere with oxidative phosphorylation. Because ammonia can be tolerated only at very low concentrations, animals that excrete ammonia need access to lots of water. Therefore, ammonia excretion is most common in aquatic species. The highly soluble ammonia molecules, which interconvert between NH_3 and NH_4^+, easily pass through membranes and are readily lost by diffusion to the surrounding water. In many invertebrates, ammonia release occurs across the whole body surface.

Urea

Although ammonia excretion works well in many aquatic species, it is much less suitable for land animals. Ammonia is so toxic that it can be safely transported through and excreted from the body only in large volumes of very dilute solutions. Most terrestrial animals and many marine species simply do not have access to sufficient water to routinely excrete ammonia. Instead, they mainly excrete a different nitrogenous waste, **urea**. In vertebrates, urea is the product of an energy-consuming metabolic cycle that combines ammonia with carbon dioxide in the liver.

The main advantages of urea for nitrogenous waste excretion are its very low toxicity and its high solubility in water. The main disadvantage is its energy cost: Animals must expend energy to produce urea from ammonia. From a bioenergetic standpoint, we would predict that animals that spend part of their lives in water and part on land would switch between excreting ammonia (thereby saving energy) and excreting urea (reducing excretory water loss). Indeed, many amphibians excrete mainly ammonia when they are aquatic tadpoles and switch largely to urea excretion when they become land-dwelling adults.

Uric Acid

Insects, land snails, and many reptiles, including birds, excrete **uric acid** as their primary nitrogenous waste **(Figure 44.7)**. Uric acid is relatively nontoxic and does not readily dissolve in water. It therefore can be excreted as a semisolid paste with very little water loss. However, uric acid is even more energetically expensive than urea, requiring considerable ATP for synthesis from ammonia.

Figure 44.7

Guano (bird excrement) is rich in uric acid.

Although they are not primarily uric acid producers, humans and some other animals generate a small amount of uric acid from metabolism. Diseases that alter this process reflect problems that can arise when a metabolic product is insoluble. For example, a genetic defect predisposes Dalmatian dogs to form uric acid stones in their bladder. In humans, adult males are particularly susceptible to *gout*, a painful joint inflammation caused by deposits of uric acid crystals. Some dinosaurs appear to have been similarly affected: Fossilized bones of *Tyrannosaurus rex* exhibit joint damage characteristic of gout.

The Influence of Evolution and Environment on Nitrogenous Wastes

EVOLUTION As a result of natural selection, the type and amount of nitrogenous waste a species produces are matched to its environment. One key factor in a habitat is the availability of water. For example, terrestrial turtles (which often live in dry areas) excrete mainly uric acid, whereas aquatic turtles excrete both urea and ammonia.

In some cases, an animal's egg is the immediate environment of relevance to the type of nitrogenous waste excreted. In an amphibian egg, which lacks a shell, ammonia or urea can simply diffuse out of the egg. Similarly, soluble wastes produced by a mammalian embryo can be carried away by the mother's blood. In the case of birds and other reptiles, however, the egg is surrounded by a shell that is permeable to gases but not to liquids. As a result, any soluble nitrogenous wastes released by the embryo would be trapped within the egg and could accumulate to dangerous levels. For this reason, using uric acid as an insoluble waste product conveys a selective advantage in reptiles. Stored within the egg as a harmless solid, the uric acid is left behind when the animal hatches.

Regardless of the type of nitrogenous waste, the amount produced is coupled to the animal's energy budget. Endotherms, which use energy at high rates, eat more food and produce more nitrogenous waste than ectotherms. The amount of nitrogenous waste is also linked to diet. Predators, which derive much of their energy from protein, excrete more nitrogen than animals that rely mainly on lipids or carbohydrates as energy sources.

Having surveyed the forms of nitrogenous waste and their interrelationship with habitat and energy consumption, we'll turn next to the processes and systems animals use to excrete these and other wastes.